

Student Name: <u>Aden Batod</u>	Final Examination Part I
Date: Monday May-04, 2026	Time: 01:00pm-03:00pm

(32) + (55) = 87

Question #01 (5pts): Select the correct answer for the Gibbs free energy G^{ID} of an ideal liquid mixture of two components with molar composition $\{x_1, x_2\}$.

☐	$G^{ID} = x_1 \bar{G}_1 + x_2 \bar{G}_2$
☐	$G^{ID} = x_1 \bar{G}_1^{ID} + x_2 \bar{G}_2^{ID}$
☒	$G^{ID} = x_1 \bar{G}_1^{ID} + x_2 \bar{G}_2^{ID} + RT(x_1 \ln x_1 + x_2 \ln x_2)$

Question #02 (5pts): Select the correct answer for the definition of the K -values of an ideal gas mixture in equilibrium with a non-ideal liquid mixture.

☐	$K_i = \frac{\hat{\phi}_i^V}{\hat{\phi}_i^L}$
☒	$K_i = \frac{P_i^{\text{sat}}}{P}$
☐	$K_i = \hat{\phi}_i^L$

Question #03 (5pts): Select the correct answer for the equation of dew point temperature of a non-ideal vapor mixture with molar composition $\{y_1, y_2, \dots, y_N\}$. Assume ideal-liquid phase mixture

$$y_i \phi_i^V = x_i \phi_i^L$$

$$x_i = y_i \frac{\phi_i^V}{\phi_i^L}$$

☐	$\sum_{i=1}^N y_i \hat{\phi}_i^V = 1$
☐	$\sum_{i=1}^N y_i \hat{\phi}_i^L = 1$
☒	$\sum_{i=1}^N y_i \frac{\hat{\phi}_i^V}{\hat{\phi}_i^L} = 1$
☐	$\sum_{i=1}^N y_i \frac{\phi_i^L}{\hat{\phi}_i^V} = 1$

benzene and toluene are 1008 mmHg and 404 mmHg respectively at 90 °C

$$\frac{1}{\text{Dew } P} = \frac{0.5}{1008} + \frac{0.5}{404} = 576.8$$

Pressure at the beginning (mmHg)	706
Pressure at the end (mmHg)	576.8

$$\text{Bubble } P = 0.5(1008) + 0.5(404) = 706$$

Question #05 (5pts): The following Equation of State $\frac{PV}{RT} = 1 + \frac{b}{V}$, where b is a constant number, describes the properties of a pure fluid. Derive a formula for the Residual Entropy as a function of (Z, B) where $B = \frac{bP}{RT}$

$$\frac{S^R}{R} =$$

Question #06 (5pts): The Equilibrium constant of a chemical reaction is independent of the pressure of the reaction mixture.

☒	True
☐	False

Question #07 (5pts): An ideal gas expands **isenthalpically** from a pressure $P_1 = 10 \text{ bar}$ to a pressure $P_2 = 1 \text{ bar}$. What is the temperature change of the gas during this process (select from the list below)

☒	$\Delta T = 0$
☐	$\Delta T < 0$
☐	$\Delta T > 0$

Question #08 (5pts): Consider the total differential $d(nG) = nVdP - nSdT + \sum_i \bar{G}_i dn_i$. Which formulas from the ones shown below are correct. You must select all that apply for full credit

☒	$\left(\frac{\partial S}{\partial P} \right)_{T,n} = - \left(\frac{\partial V}{\partial T} \right)_{P,n}$
☒	$\left(\frac{\partial \bar{G}_i}{\partial P} \right)_{T,n_j \neq i} = \bar{V}_i$
☒	$\left(\frac{\partial \bar{G}_i}{\partial T} \right)_{P,n_j \neq i} = -\bar{S}_i$

Question #09 (5pts): The fugacity coefficient of a component inside an ideal-liquid mixture is equal to one.

☐	True
☒	False

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Question #10 (5pts): A single VLE experimental data point is available at the bubble point temperature $T_{\text{BUBBLE}} = 35^\circ\text{C}$ under pressure $P = 76.719 \text{ kPa}$ with composition $[x_1, y_1] = [0.1665, 0.0832]$ for a binary mixture. The saturated vapor pressures of the two components are $P_1^{\text{sat}} = 49.624 \text{ kPa}$ and $P_2^{\text{sat}} = 85.265 \text{ kPa}$ at the bubble point temperature. What is the value of the coefficient A in the Margules 2-suffix equation for the Excess Gibbs Free Energy. Assume ideal gas mixture for the vapor phase.

Value of A	1.55
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$$T_{\text{Bubble}} = 35^\circ\text{C}$$

$$P = 0.76719 \text{ Bar}$$

$$x_1 = 0.1665$$

$$y_1 = 0.0832$$

$$P_1^{\text{sat}} = 0.49624 \text{ bar}$$

$$P_2^{\text{sat}} = 0.85265 \text{ bar}$$

$$y_1 P_b = x_1 y_1 P^{\text{sat}}$$

$$0.0832(0.76719) = 0.1665 \gamma_1 P_1^{\text{sat}}$$

$$0.06382272 = 0.0826296 \gamma_1$$

$$\gamma_1 = 0.7724$$

$$A = \frac{\ln(0.7724)}{0.1665}$$

Student Name: <u>Aden Rataul</u>	Final Examination Part II
Date: Monday May-04, 2026	Time: 01:00pm-03:00pm

Problem #01 (15 pts): A gas mixture containing 70 mol% Hydrogen, 20 mol% Nitrogen and 10 mol% inert gas Argon is passed over a catalyst for production of Ammonia. If the pressure of the reaction mixture is 50 atm and the final temperature 400 °C, calculate the percentage of Hydrogen converted to Ammonia and the mole fraction of Ammonia in the reaction mixture. Assume the gas mixture follows the Peng-Robinson Equation of State with Binary Interaction Parameters all set equal to zero.

H ₂ Conversion (%)	23.96
NH ₃ mole fraction	0.1259



$$\nu_{\text{H}_2} = -3 \quad \nu_{\text{N}_2} = -1 \quad \nu_{\text{NH}_3} = +2 \quad \nu_{\text{Ar}} = 0$$

1 mol basis:

$$n_{\text{H}_2,0} = 0.7 \quad n_{\text{N}_2,0} = 0.2 \quad n_{\text{NH}_3,0} = 0 \quad n_{\text{Ar},0} = 0.1$$

$\xi = \text{guess}$

$$n_{\text{H}_2} = 0.7 - 3\xi$$

$$n_{\text{N}_2} = 0.2 - \xi$$

$$n_{\text{NH}_3} = 0 + 2\xi$$

$$n_{\text{Ar}} = 0.1 + 0\xi$$

mole fraction: $y_i = \frac{n_i}{n_{\text{tot}}}$

$k = k_{\text{calc}}$ by using solver and changing $\xi \rightarrow 0.0559$

$$X_{\text{H}_2} = \frac{\text{moles H}_2 \text{ reacted}}{\text{initial moles H}_2} \cdot 100 = \frac{3(0.0559)}{0.7} \cdot 100 = \boxed{23.96\%}$$

$$y_{\text{NH}_3} = \frac{n_{\text{NH}_3}}{n_{\text{tot}}} = \boxed{0.1259}$$

properties of Nitrogen:

$$P + \frac{a}{TV^2} = \frac{RT}{V-b} \quad \left| \quad a = \lambda_1 \frac{R^2 T_c^3}{P_c} \quad \right| \quad b = \lambda_2 \frac{RT_c}{P_c}$$

Calculate the values of the numerical constants $\{\lambda_1, \lambda_2\}$ and the value of the critical compressibility factor Z_c . Derive an expression for the calculation of the residual enthalpy and calculate its value at $T=50^\circ\text{C}$ and pressure $P=650\text{ atm}$. **Hint:** Use an initial guess of $Z^{(0)} = 2$ for the calculation of the residual enthalpy

λ_1	0.421875
λ_2	0.125
Z_c	0.375
$\frac{H^R}{RT}$	0.759

$$\frac{\partial P}{\partial V} = -\frac{RT}{(V-b)^2} + \frac{2a}{TV^3}$$

$$\frac{\partial^2 P}{\partial V^2} = \frac{2RT}{(V-b)^3} - \frac{6a}{TV^4}$$

$$\frac{RT_c}{(V_c-b)^2} = \frac{2a}{T_c V_c^3}$$

$$\frac{RT_c}{(V_c-b)^3} = \frac{3a}{T_c V_c^4}$$

Problem #03 (15 pts): The mixture of Methanol (Component #1), Acetone (Component #2) and Cyclohexane (Component #3) at $T = 101^\circ\text{C}$ forms an azeotrope $\{y_1 = x_1, y_2 = x_2 \text{ and } y_3 = x_3\}$. Determine the pressure and composition that is predicted for the azeotrope between methanol, acetone and cyclohexane by using the NRTL method with binary interaction parameters, $\{A_{ij}, A_{ji}, B_{ij}, B_{ji}, C\}$ shown in the table and assuming that the vapor phase follows the **Redlich-Kwong** Equation of State with all BIPs set equal to zero. **Poynting Factor=1**

BIPs		CH ₄ O C ₃ H ₆ O	CH ₄ O C ₆ H ₁₂	C ₃ H ₆ O C ₆ H ₁₂
KA _{ij}		0.0000	0.0000	0.0000
KB _{ij}				
KC _{ij}				
A _{ij}		0.0000	1.3869	-0.1061
A _{ji}		0.0000	-4.6753	-3.5142
B _{ij}		114.1347	224.7601	359.6793
B _{ji}		101.8859	2277.7942	1485.3369
C		0.3000	0.4300	0.4700

Pressure (bar)	7.7526
Methanol mole fraction	0.5876
Acetone mole fraction	0.1184
Cyclohexane mole fraction	0.2930

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$$k_i = \frac{y_i}{x_i} = 1$$

$$P, \text{ bar} = \text{guess}$$

$$x_i \text{ and } y_i = \text{guess}$$

$$x_i y_i P_i^{\text{sat}} \phi_i^{\text{sat}} = y_i \phi_i^V P$$

$$\frac{y_i}{x_i} = \frac{P_i^{\text{sat}} \phi_i^{\text{sat}}}{\phi_i^V P}$$

$$k_i = \frac{P_i^{\text{sat}} \phi_i^{\text{sat}}}{\phi_i^V P}$$

$$\text{Objective} = (k_1 - 1)^2 + (k_2 - 1)^2 + (k_3 - 1)^2$$

minimize the objective with solver
changing P, x_i

Problem #04 (15 pts): A liquid hydrocarbon mixture of Methane (Comp #1), Ethane (Comp #2) and Propane (Comp #3) at temperature $T_1 = -100\text{ }^{\circ}\text{C}$ and pressure $P_1 = 64\text{ bar}$ with molar composition $z = [0.80, 0.08, 0.12]$ expands **adiabatically** down to a pressure of $P_2 = 10\text{ bar}$ in a liquid turbine. If the temperature at the outlet of the liquid turbine is $T_2 = -118.5\text{ }^{\circ}\text{C}$ calculate the power generated in kW. The flow rate of the hydrocarbon mixture is 10,000 kmol/hr. Also report the calculated vapor fraction at the turbine outlet and the Lost Work in kW. Assume that both phases of the hydrocarbon mixture follow the Peng-Robinson Equation of State with all BIPs set equal to zero. Take the temperature of the infinite environment to be equal to $T_0 = 300\text{ K}$

Power (kW)	78 6.07
Vapor fraction at the turbine outlet	0.13124
Lost Work (kW)	35.68

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$$\dot{W}_{out} = \dot{n} (h_1 - h_2)$$

$$\dot{S}_{gen} = \dot{n} (s_2 - s_1)$$